

```
the_boss@LAPTOP-FGPPB110:~$ mkdir OSSP
the_boss@LAPTOP-FGPPB110:~$ cd OSSP
the_boss@LAPTOP-FGPPB110:~/OSSP$ mkdir src include obj bin docs tests scripts assets
mkdir: cannot create directory 'OSSP': File exists
mkdir: cannot create directory 'src': File exists
mkdir: cannot create directory 'include': File exists
mkdir: cannot create directory 'obj': File exists
mkdir: cannot create directory 'bin': File exists
mkdir: cannot create directory 'docs': File exists
mkdir: cannot create directory 'tests': File exists
mkdir: cannot create directory 'scripts': File exists
mkdir: cannot create directory 'assets': File exists
the_boss@LAPTOP-FGPPB110:~/OSSP$ ls
LICENSE  README.md  bin  include  scripts  tests
Makefile  assets     docs  obj      src
the_boss@LAPTOP-FGPPB110:~/OSSP$ ls -l
total 32
-rw-r--r-- 1 the_boss the_boss  0 Aug  1 07:59 LICENSE
-rw-r--r-- 1 the_boss the_boss  0 Aug  1 07:59 Makefile
-rw-r--r-- 1 the_boss the_boss  0 Aug  1 07:59 README.md
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 assets
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:58 bin
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 docs
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 include
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:58 obj
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 scripts
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 src
drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 tests
the_boss@LAPTOP-FGPPB110:~/OSSP$ find . -maxdepth 3
./obj
the_boss@LAPTOP-FGPPB110:~/OSSP$
```

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drwxr-xr-x 2 the_boss the_boss 4096 Aug  1 07:59 tests
the_boss@LAPTOP-FGPPB110:~/OSSP$ find . -maxdepth 3
./obj
the_boss@LAPTOP-FGPPB110:~/OSSP$
```

```
./ .git/info
./ .git/info/exclude
./ README.md
./ assets
./ assets/architecture.png
./ scripts
./ scripts/run.sh
./ include
./ include/shell.h
./ include/executor.h
./ include/parser.h
./ include/builtin.h
./ LICENSE
./ .gitignore
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ █
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ touch tests/test_parser.c
the_boss@LAPTOP-FGPPB110:~/OSSP$ touch scripts/run.sh as;
the_boss@LAPTOP-FGPPB110:~/OSSP$ find . -maxdepth 3 -type
./src/executor.c
./src/parser.c
./src/builtin.c
./src/shell.c
./src/main.c
./Makefile
./docs/report.pdf
./docs/design.md
./tests/test_executor.c
./tests/test_parser.c
./.git/description
./.git/config
./.git/hooks/applypatch-msg.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/hooks/post-update.sample
./.git/hooks/push-to-checkout.sample
./.git/hooks/pre-applypatch.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/update.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/pre-commit.sample
./.git/hooks/commit-msg.sample
./.git/hooks/pre-push.sample
./.git/hooks/pre-receive.sample
./.git/hooks/prepare-commit-msg.sample
./.git/HEAD
./.git/info/exclude
./README.md
./assets/architecture.png
./scripts/run.sh
./include/shell.h
./include/executor.h
./include/parser.h
./include/builtin.h
./LICENSE
./.gitignore
the_boss@LAPTOP-FGPPB110:~/OSSP$ █
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ git add .
the_boss@LAPTOP-FGPPB110:~/OSSP$ git status
On branch master
```

No commits yet

Changes to be committed:

(use "git rm --cached <file>..." to unstage)

```
new file:   .gitignore
new file:   LICENSE
new file:   Makefile
new file:   README.md
new file:   assets/architecture.png
new file:   docs/design.md
new file:   docs/report.pdf
new file:   include/builtin.h
new file:   include/executor.h
new file:   include/parser.h
new file:   include/shell.h
new file:   scripts/run.sh
new file:   src/builtin.c
new file:   src/executor.c
new file:   src/main.c
new file:   src/parser.c
new file:   src/shell.c
new file:   tests/test_executor.c
new file:   tests/test_parser.c
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ git commit -m "Initial OSSP project structure"
Author identity unknown
the_boss@LAPTOP-FGPPB110:~/OSSP$ █
```



```
fatal: empty ident name (for <the_boss@LAPTOP-FGPPB110.
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ git status
```

```
On branch master
```

```
No commits yet
```

```
Changes to be committed:
```

```
(use "git rm --cached <file>..." to unstage)
```

```
new file:   .gitignore
new file:   LICENSE
new file:   Makefile
new file:   README.md
new file:   assets/architecture.png
new file:   docs/design.md
new file:   docs/report.pdf
new file:   include/builtin.h
new file:   include/executor.h
new file:   include/parser.h
new file:   include/shell.h
new file:   scripts/run.sh
new file:   src/builtin.c
new file:   src/executor.c
new file:   src/main.c
new file:   src/parser.c
new file:   src/shell.c
new file:   tests/test_executor.c
new file:   tests/test_parser.c
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ █
```

```
fatal: empty ident name (for <the_boss@LAPTOP-FGPPB110.  
the_boss@LAPTOP-FGPPB110:~/OSSP$ git status  
On branch master
```

```
No commits yet
```

```
Changes to be committed:
```

```
(use "git rm --cached <file>..." to unstage)
```

```
new file:   .gitignore  
new file:   LICENSE  
new file:   Makefile  
new file:   README.md  
new file:   assets/architecture.png  
new file:   docs/design.md  
new file:   docs/report.pdf  
new file:   include/builtin.h  
new file:   include/executor.h  
new file:   include/parser.h  
new file:   include/shell.h  
new file:   scripts/run.sh  
new file:   src/builtin.c  
new file:   src/executor.c  
new file:   src/main.c  
new file:   src/parser.c  
new file:   src/shell.c  
new file:   tests/test_executor.c  
new file:   tests/test_parser.c
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ █
```

```
GNU nano 7.2                                                                    fork
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
the_boss@LAPTOP-FGPPB110:~/OSSP$ nothing to commit, working tree clean
nothing: command not found
the_boss@LAPTOP-FGPPB110:~/OSSP$ ^C
the_boss@LAPTOP-FGPPB110:~/OSSP$ nano fork_example.c
the_boss@LAPTOP-FGPPB110:~/OSSP$ gcc fork_example.c -o fork_example
the_boss@LAPTOP-FGPPB110:~/OSSP$ ./fork_example
This is the Parent Process.
Parent PID = 882
Child PID = 883
This is the Child Process.
Child PID = 883
the_boss@LAPTOP-FGPPB110:~/OSSP$ █
```